Click or calculate? Comparing paper and AI-supported homework in high-school projectile motion

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Abstract

Generative AI is changing how students do homework, but it is unclear whether AI help supports learning or replaces it. This study compared three homework formats for the same horizontal projectile motion problem with 292 Grade 11 students in China: traditional paper homework, and an online page with an interactive simulation combined either with a standard AI chatbot or with a custom AI tutor that guided students step by step and adapted its help to their needs. One week later, students took a post-test without AI that contained a near-transfer and a far-transfer problem. The paper homework group performed best on both problems and the standard AI group worst, with the custom AI tutor group in between (Hedges' g between the paper and custom AI tutor groups: 0.39 on the near-transfer and 0.47 on the far-transfer problem). Students' answer sheets showed that the custom AI tutor group drew the two limiting trajectories far more often than the other groups (47%, compared with 3% and 11%), including in the new context, and came close to the paper group on a qualitative transfer question, but often could not complete the calculation. The custom AI tutor thus appears to have helped students see the structure of the problem, but not work it out. The results suggest that, as long as physics teaching expects students to solve problems quantitatively on their own, AI tutors for homework should leave the calculation to the student.

Keywords: generative AI; AI tutor; homework; projectile motion; transfer; high-school physics

Supplementary material: Appendices A and B (the AI prompts), the post-test scores and the blurred answer sheets are available at https://phylab.uk/research/Click_or_Calculate/

Introduction

Homework is where students practise solving problems on their own. Generative AI has changed this practice: students can now turn to an AI as soon as they get stuck, or even before they have tried to think the problem through, and the AI will usually explain the next step or provide the whole solution. Whether this help supports learning or replaces it has become a pressing question for physics teachers.

Recent studies point in both directions. In a field experiment with nearly 1,000 high-school mathematics students, Bastani et al. (2025) found that access to GPT-4 through a standard chat interface improved performance on practice problems by 48%, but lowered performance on a later exam without AI by 17%. A version with prompts designed to safeguard learning, which instructed the AI to give hints rather than answers, improved practice performance by 127%, and its students performed no worse than the control group on the exam, though also no better. In physics, Krupp et al. (2024) observed that university students using ChatGPT pasted the problem directly into the chatbot in 42% of their queries and that nearly half of the solutions they produced with ChatGPT were incorrect but believed by the students to be correct. In a scientific inquiry task, Stadler et al. (2024) found that university students using a large language model experienced less mental effort but produced lower-quality reasoning than those using a search engine. In a randomised experiment with 122 secondary-school students aged 14 to 16, Blasco and Charisi (2025) found that a Socratic chatbot increased engagement but did not significantly improve learning, and that students struggled to apply what they had learned to a new question once AI assistance was removed. On the other hand, Kestin et al. (2025) showed that an AI tutor carefully designed around research-based teaching practices helped students in an introductory university physics course learn more than in-class active learning did.

These findings suggest that how AI support is designed matters as much as whether it is offered at all, and so does how students use it: Lehmann et al. (2024) found that students who used a language model to generate solutions to exercises

covered more topics but understood each less well, whereas those who used it to ask for explanations understood more. Step-based tutoring systems, which respond to each step of a solution rather than only to the final answer, have been found to be nearly as effective as human tutors (VanLehn, 2011). Yet any form of help faces what Koedinger and Aleven (2007) called the assistance dilemma: giving information makes practice easier, but it may also take away work that students need to do themselves in order to learn. A custom AI tutor that breaks a problem into small steps, offers options and corrects mistakes immediately may therefore improve performance during practice and still leave students unprepared to solve a problem on their own.

It is less clear how AI support compares with the homework that most secondary-school students still do: solving a problem on paper and then checking the work against a written solution. The closest comparison is the control condition of Bastani et al. (2025), in which students practised with their textbook and notes, and the teacher reviewed the correct answers only after they had submitted their work. That study, however, was in mathematics, and its safeguarded tutor remained passive, answering students' questions without guiding them through the problem, a limitation the authors themselves noted. Kestin et al. (2025) compared their AI tutor with in-class active learning rather than with homework. An earlier comparison of web-based and paper homework in physics found no significant difference in performance (Bonham et al., 2003), but the online homework in that study consisted of standard end-of-chapter exercises and offered no conversational help. Moreover, most studies report only scores. Little is known about what students actually take away from AI-supported homework: whether they learn to represent the problem correctly, to carry out the calculation, or both, and whether this learning transfers to a new context.

This study compared three homework formats for the same horizontal projectile motion problem with 292 Grade 11 students in China: traditional paper homework with a written solution, and an online page with an interactive simulation combined either with a standard AI chatbot or with a custom AI tutor that guided students step by step and adapted its help to their needs. One week later, all students took a post-test without AI that contained a near-transfer and a far-transfer problem. The study addressed two research questions:

1. When AI is no longer available, how do students who completed traditional paper homework, homework with a standard AI, or homework with a custom AI tutor perform on near-transfer and far-transfer problems?
2. What does students' written work reveal about whether each homework format helped students represent the problem and carry out the calculation?

Method

This study compared three groups of students (N = 292) with similar academic backgrounds. Each group completed the same physics homework problem in a different format, and all groups then took the same post-test.

Participants

A total of 292 students from six parallel Grade 11 classes took part in the study. Throughout the paper, including the comparison of prior examination scores below, only students who completed both the homework and the post-test were included. When they entered high school, students were placed in classes according to a weighted average of their scores on the senior high school entrance examination and a school placement test, in a serpentine order: with n classes, the students ranked 1 to n by this score were placed in classes 1 to n respectively, those ranked $n + 1$ to $2n$ in classes n to 1, and so on. After more than a year of study, however, the average scores of the classes had drifted slightly apart. I therefore combined the six classes into three groups of two classes each, choosing the pairs so that the three groups were as similar as possible before the study.

Figure 1 shows the mean scores of the three groups on six school physics examinations, five held before the homework and one after it. The groups were very similar on every examination; one-way ANOVAs found no significant difference between them on any examination (all $p > 0.17$).

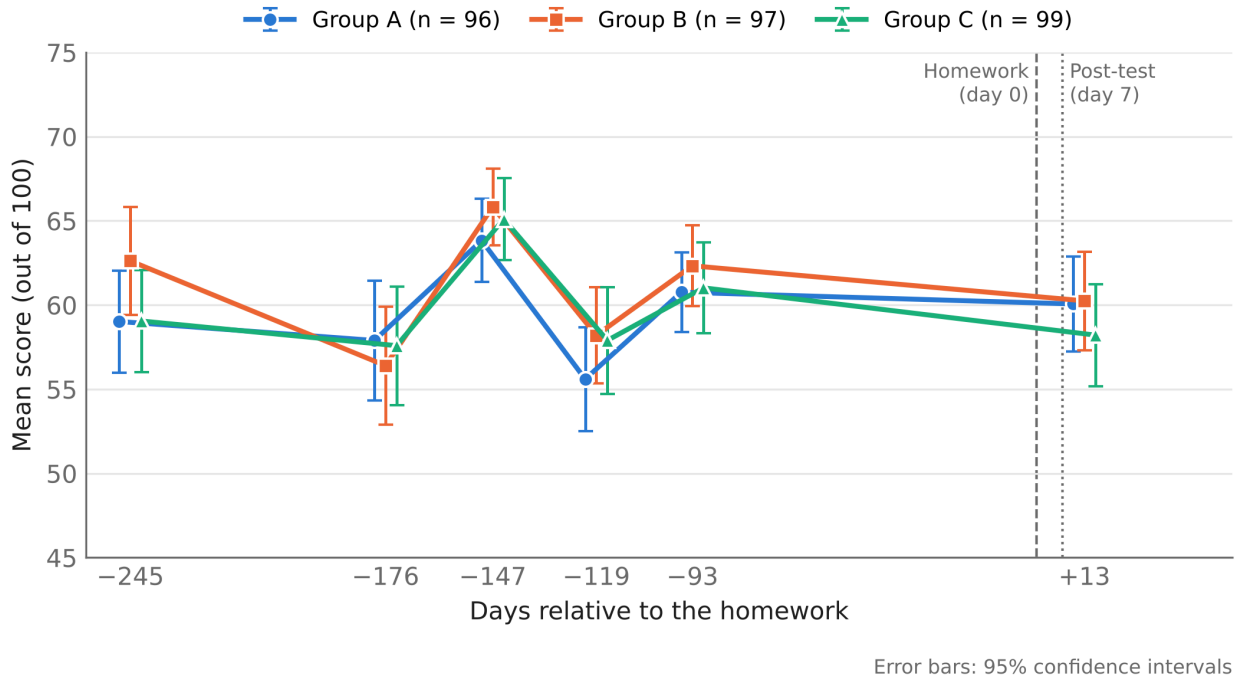
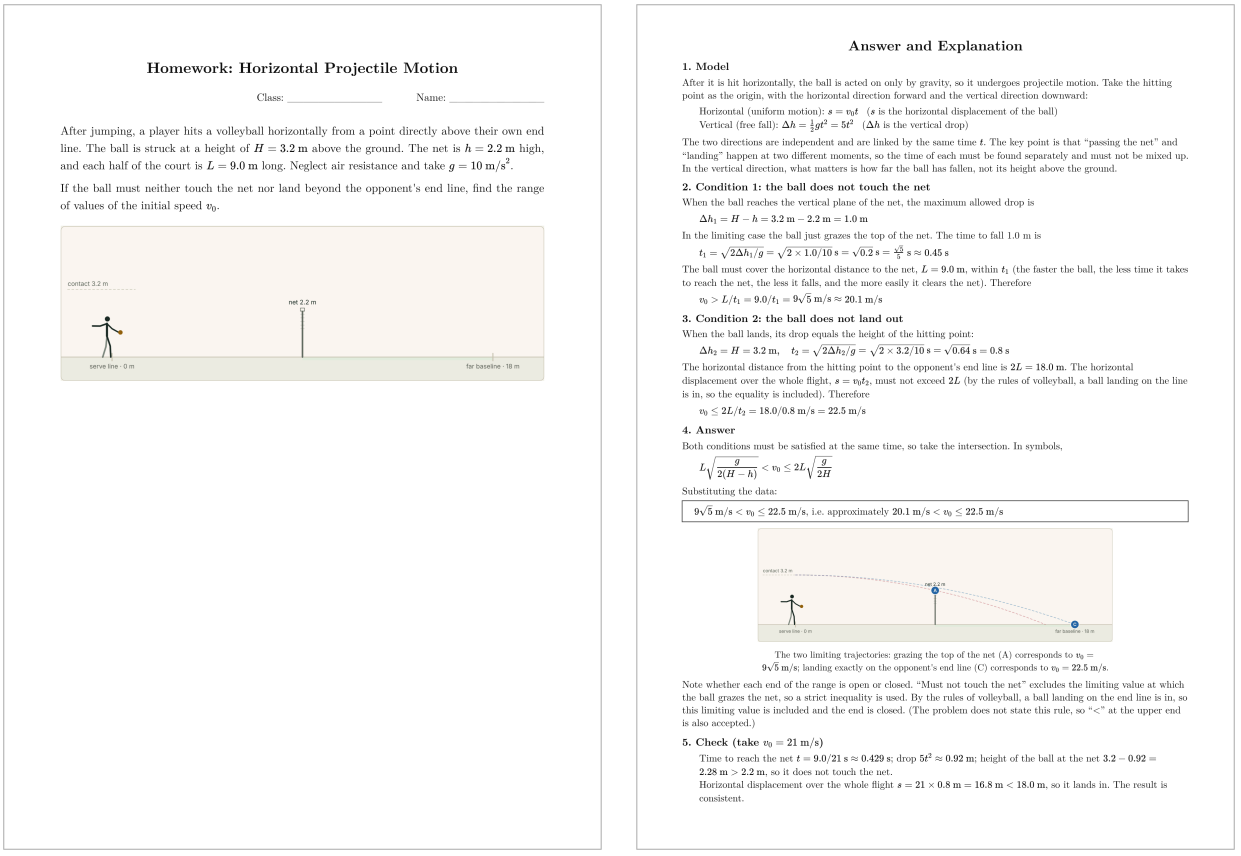


Figure 1. Mean scores of Group A (n = 96), Group B (n = 97) and Group C (n = 99) on six school physics examinations, plotted against the number of days relative to the homework. The dashed line marks the homework and the dotted line marks the post-test (day 7), which is not one of the six examinations. Error bars show 95% confidence intervals.

Homework conditions

The three groups of students were given three types of homework: traditional paper homework, online homework with a standard AI, and online homework with a custom AI tutor. All three groups worked on the same problem. Students saw all materials in Chinese; the problem, the solution and the student-facing pages are shown here in English translation.

Group A: traditional paper homework. As in their everyday homework, students received one page with the problem and one page with the answer and explanations (Figure 2). They were asked to solve the problem first and then check and correct their work against the answer. Because they have long been taught good homework habits, most students make a serious attempt at the problem, even if they cannot complete it, before looking at the answer.



Homework: Horizontal Projectile Motion

Class: _____ Name: _____

After jumping, a player hits a volleyball horizontally from a point directly above their own end line. The ball is struck at a height of $H = 3.2\ \text{m}$ above the ground. The net is $h = 2.2\ \text{m}$ high, and each half of the court is $L = 9.0\ \text{m}$ long. Neglect air resistance and take $g = 10\ \text{m/s}^2$.

If the ball must neither touch the net nor land beyond the opponent's end line, find the range of values of the initial speed v_0.

Answer and Explanation

1. Model

After it is hit horizontally, the ball is acted on only by gravity, so it undergoes projectile motion. Take the hitting point as the origin, with the horizontal direction forward and the vertical direction downward:

Horizontal (uniform motion): $s = v_0 t$ (s is the horizontal displacement of the ball)

Vertical (free fall): $\Delta h = \frac{1}{2} g t^2 = 5t^2$ (Δh is the vertical drop)

The two directions are independent and are linked by the same time t. The key point is that "passing the net" and "landing" happen at two different moments, so the time of each must be found separately and must not be mixed up. In the vertical direction, what matters is how far the ball has fallen, not its height above the ground.

2. Condition 1: the ball does not touch the net

When the ball reaches the vertical plane of the net, the maximum allowed drop is

$$\Delta h_1 = H - h = 3.2\ \text{m} - 2.2\ \text{m} = 1.0\ \text{m}$$

In the limiting case the ball just grazes the top of the net. The time to fall 1.0 m is

$$t_1 = \sqrt{2\Delta h_1 / g} = \sqrt{2 \times 1.0/10}\ \text{s} = \sqrt{0.2}\ \text{s} = \frac{\sqrt{2}}{5}\ \text{s} \approx 0.45\ \text{s}$$

The ball must cover the horizontal distance to the net, $L = 9.0\ \text{m}$, within t_1 (the faster the ball, the less time it takes to reach the net, the less it falls, and the more easily it clears the net). Therefore

$$v_0 > L/t_1 = 9.0/t_1 = 9\sqrt{5}\ \text{m/s} \approx 20.1\ \text{m/s}$$

3. Condition 2: the ball does not land out

When the ball lands, its drop equals the height of the hitting point:

$$\Delta h_2 = H = 3.2\ \text{m}, \quad t_2 = \sqrt{2\Delta h_2 / g} = \sqrt{2 \times 3.2/10}\ \text{s} = \sqrt{0.64}\ \text{s} = 0.8\ \text{s}$$

The horizontal distance from the hitting point to the opponent's end line is $2L = 18.0\ \text{m}$. The horizontal displacement over the whole flight, $s = v_0 t_2$, must not exceed $2L$ (by the rules of volleyball, a ball landing on the line is in, so the equality is included). Therefore

$$v_0 \le 2L/t_2 = 18.0/0.8\ \text{m/s} = 22.5\ \text{m/s}$$

4. Answer

Both conditions must be satisfied at the same time, so take the intersection. In symbols,

$$L\sqrt{\frac{g}{2(H-h)}} < v_0 \le 2L\sqrt{\frac{g}{2H}}$$

Substituting the data:

$$9\sqrt{5}\ \text{m/s} < v_0 \le 22.5\ \text{m/s}, \text{ i.e. approximately } 20.1\ \text{m/s} < v_0 \le 22.5\ \text{m/s}$$

The two limiting trajectories: grazing the top of the net (A) corresponds to $v_0 = 9\sqrt{5}\ \text{m/s}$; landing exactly on the opponent's end line (C) corresponds to $v_0 = 22.5\ \text{m/s}$.

Note whether each end of the range is open or closed. "Must not touch the net" excludes the limiting value at which the ball grazes the net, so a strict inequality is used. By the rules of volleyball, a ball landing on the end line is in, so this limiting value is included and the end is closed. (The problem does not state this rule, so "<" at the upper end is also accepted.)

5. Check (take $v_0 = 21\ \text{m/s}$)

Time to reach the net $t = 9.0/21\ \text{s} \approx 0.429\ \text{s}$; drop $5t^2 \approx 0.92\ \text{m}$; height of the ball at the net $3.2 - 0.92 = 2.28\ \text{m} > 2.2\ \text{m}$, so it does not touch the net.

Horizontal displacement over the whole flight $s = 21 \times 0.8\ \text{m} = 16.8\ \text{m} < 18.0\ \text{m}$, so it lands in. The result is consistent.

Figure 2. Group A homework: the question page (left) and the answer page with explanations (right).

Groups B and C completed the homework on a computer, using the same online page (Figure 3). In both groups, a simulation computed the trajectory of each serve at the speed the student chose and judged whether the ball hit the net, landed in, or went long. Interactive simulations of this kind can focus students' attention on the underlying physics and let them explore by changing parameters (Wieman et al., 2008). The two groups differed only in the AI that helped them, which appeared in the right-hand panel of the page.

Group B: online homework with a standard AI. The right-hand panel contained a chat box connected to DeepSeek with a minimal prompt that contained only the problem statement (Appendix A), so that the AI behaved like an ordinary chatbot.

Group C: online homework with a custom AI tutor. The right-hand panel contained a custom tutor with two parts. A scripted Coach used the outcome of the student's own serve to decide where instruction began, and guided the student step by step to the two boundaries of the speed range: for each boundary, the student identified the boundary point, watched a limiting serve, and then found the horizontal distance, the vertical fall, the fall time and the speed. Whole-number quantities were typed in; other quantities were chosen from options, and each wrong option corresponded to a common mistake and triggered specific feedback. In the terms of VanLehn (2011), the scripted Coach was thus a step-based tutor. At any point, students could also ask free-form questions of an AI Coach, which was connected to the same DeepSeek model as in Group B. With every question, the AI Coach received the complete solution, the current state of the simulation (the chosen speed, the outcome of the latest serve, the height of the ball at the net, the landing point and the number of attempts) and the student's position in the scripted sequence, so that it could analyse the student's own serves and progress and tailor its reply. It was instructed to give help step by step rather than the full solution at once, and to answer the question asked before returning the student to the scripted sequence. The full prompt is given in Appendix B.

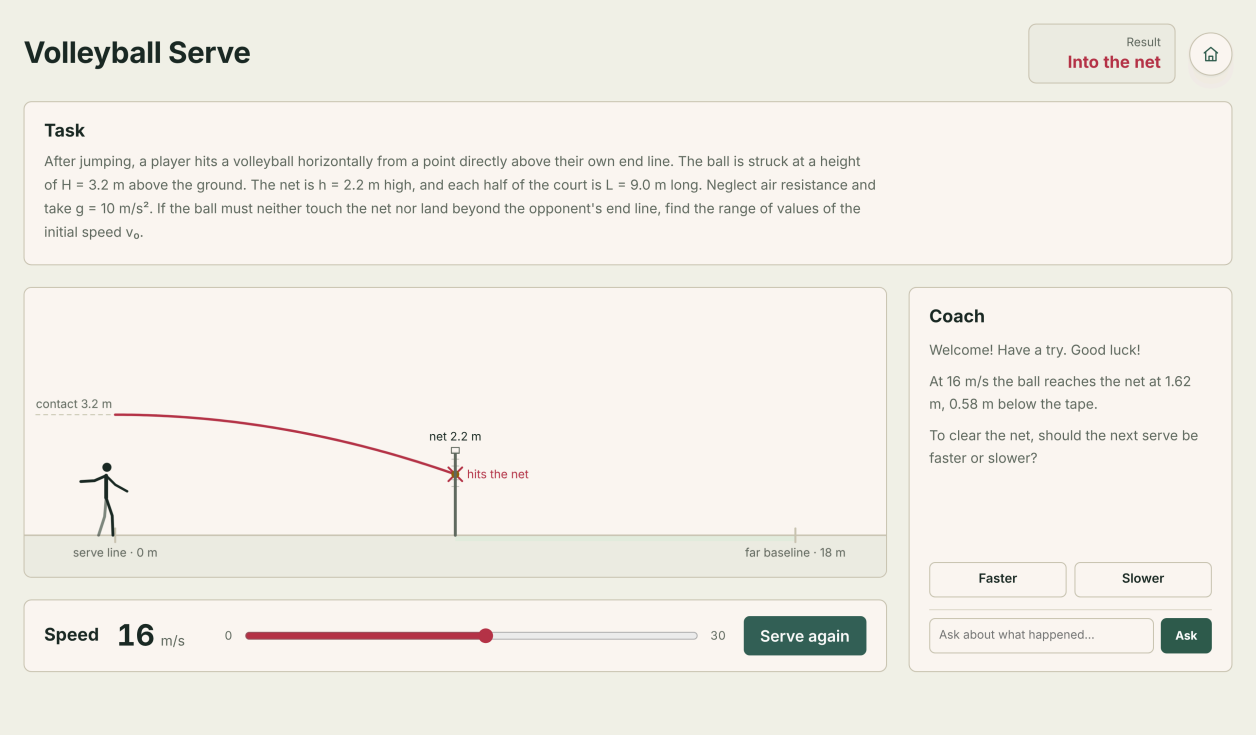


Figure 3. The online homework page as used by Group C (phylab.uk/labs/volleyball-serve; the English interface is shown, whereas students used the Chinese interface). The student has served at 16 m/s and the ball has hit the net; the scripted Coach asks whether the next serve should be faster or slower. The box at the bottom right lets students ask the AI Coach free-form questions at any time. Group B used the same page with the same simulation, but the right-hand panel contained only a chat box connected to a standard AI.

All three groups received the homework on the same day, one week before the post-test. From my teaching experience, I estimated that the homework would take about 20 minutes. Group A did the homework during an evening self-study session. For Groups B and C, 20 minutes was set aside for the homework at the end of a regular Python programming lesson in the computer room. Each student in Groups B and C was given a sheet of scratch paper for calculations. The homework sheets of Group A and the scratch paper of Groups B and C were collected for analysis.

Post-test

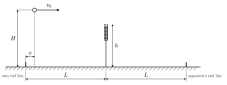
One week after the homework, all three groups took a post-test under supervision. Students were not allowed to use AI or to ask others for help, and the time limit was 30 minutes. The post-test contained two problems (Figure 4). Problem 1 was a near-transfer problem: it used the same volleyball context as the homework, with different numbers and one additional condition, the player's forward jump of x = 1.0 m before hitting the ball. Problem 2 was a far-transfer problem: it placed horizontal projectile motion in a new context, a parcel-sorting line, and asked students to apply what they had learned in the homework. Transfer can be near or far along several dimensions (Barnett & Ceci, 2002); the two problems differed mainly in how far their context was from that of the homework. Problem 2 had two parts: Problem 2a asked for the range of belt speeds for which a parcel lands in Box 2, and Problem 2b asked students to propose improvements that would raise the chance of a parcel landing in Box 2 when the belt speed fluctuates.

Test: Horizontal Projectile Motion

Class: _____ Name: _____

1. A player takes off from behind their own end line and moves forward in the air. When the hitting point is $x = 1.0\ \text{m}$ in front of their own end line, the player hits the volleyball horizontally, perpendicular to the net. The hitting point is $H = 3.2\ \text{m}$ above the ground. The net is $h = 2.4\ \text{m}$ high, and each half of the court is $L = 9.0\ \text{m}$ long. Treat the ball as a point particle, neglect air resistance and take $g = 10\ \text{m/s}^2$.

If the ball must neither touch the net nor land beyond the opponent's end line, find the range of values of the initial speed v_0.



(not to scale)

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2. On a parcel sorting line, a sufficiently long horizontal conveyor belt runs at a constant speed v. When a parcel reaches the end of the belt, it leaves the belt horizontally with speed v. The end of the belt is $H = 1.2\ \text{m}$ above the ground. In front of the end of the belt, three identical open boxes, numbered 1, 2 and 3, stand in a row on the ground. Each box is $h = 0.4\ \text{m}$ high and $L = 0.6\ \text{m}$ wide along the direction of motion. Adjacent boxes are separated by a gap of $d = 0.2\ \text{m}$, and the horizontal distance from the near edge of Box 1 (the edge closer to the belt) to the point directly below the end of the belt is also d. Treat the parcel as a point particle, neglect air resistance and the thickness of the box walls, and take $g = 10\ \text{m/s}^2$.

(1) For the parcel to land in Box 2, find the range of values of the belt speed v.

(2) In actual operation, the belt speed may fluctuate slightly around its set value. To increase the success rate of parcels landing in Box 2, propose as many different improvements as you can and briefly explain the reason for each. You may include sketches.

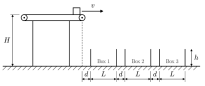


Figure 4. The post-test paper. Problem 1 (volleyball) is the near-transfer problem and Problem 2 (parcel sorting) is the far-transfer problem.

Scoring and coding

Problem 1 and Problem 2a were scored against key steps, using the same rules for all three groups (Table 1). For each boundary of the speed range, one point was given for each of the vertical drop, the fall time, the horizontal distance and the boundary speed, and one more point was given for combining the two boundaries into the correct interval. Problem 1 was therefore worth 9 points. In Problem 2a, both boundaries share the same vertical drop and fall time, which were credited only once, so Problem 2a was worth 7 points.

Table 1. Scoring points for Problem 1 and Problem 2a.

Scoring point	Problem 1 (volleyball)	Points	Problem 2a (parcel sorting)	Points
Vertical drop	Net: H − h = 0.8 m; baseline: H = 3.2 m	2	H − h = 0.8 m, shared by both boundaries	1
Fall time	Net: t_1 = 0.4 s; baseline: t_2 = 0.8 s	2	t = 0.4 s, shared by both boundaries	1
Horizontal distance	Net: L − x = 8 m; baseline: 2L − x = 17 m	2	Near edge of Box 2: 2d + L = 1.0 m; far edge: 2d + 2L = 1.6 m	2
Boundary speed	$v_0 > 20$ m/s; $v_0 \le 21.25$ m/s	2	v > 2.5 m/s; v < 4.0 m/s	2
Interval	20 m/s < $v_0 \le 21.25$ m/s	1	2.5 m/s < v < 4.0 m/s	1
Total		9		7

The following rules applied to both problems. A correct symbolic expression, such as H − h or L − x, earned the point without numerical substitution. When a later quantity was correct, skipped intermediate steps were credited, and equivalent methods, such as using the trajectory equation, earned full credit. Either a strict or a non-strict inequality was accepted at each end of the interval, but the interval point required both endpoints and their intersection. No points were deducted for units or presentation, and numerical results correct to 0.1 were accepted. A sketch without any equation earned no points.

Problem 2b was not included in the score. Instead, I counted the number of feasible improvements each student proposed, using the same criterion for all three groups. A proposal was counted as feasible if its physics pointed in the right direction and it could reasonably be understood as raising the chance that a parcel lands in Box 2, even if the accompanying conditions or explanation were incomplete. Proposals in a clearly wrong direction were not counted, and the same proposal was counted only once. Feasible proposals included, for example, shortening the flight time or modifying the shape of Box 2.

I scored all answer sheets. A second physics teacher independently scored a random 20% of the answer sheets (n = 58) using the same rules. The two scorers gave identical scores on 54 of these answer sheets (93%), and the remaining scores differed by no more than 1 point. Scoring was blind to group membership. For Problem 2b, because I had given the second scorer a list of feasible improvements, the two scorers' counts were identical.

For both problems, I also coded the trajectories that each student drew on the answer sheet into four categories: no trajectory; one trajectory (including several trajectories that did not include both limiting ones); two limiting trajectories; and incorrect, for a trajectory that was physically wrong, such as one starting upwards, a straight or broken line, or a faster ball drawn below a slower one. For the homework, I also recorded whether the scratch paper handed in by each student in Groups B and C was blank.

Data analysis

Differences between groups on the post-test were expressed as Hedges' g with 95% confidence intervals. Within Group C, students who drew the two limiting trajectories were compared with those who drew none using Mann–Whitney U tests, because many scores were 0 or full marks.

Results

Table 2 summarises the post-test results. Group A performed best and Group B worst on every measure, with Group C in between. The first two subsections below address the first research question, and the last two address the second.

Table 2. Post-test results: mean (SD) for each group, and the standardised difference between groups (Hedges' g with 95% confidence interval; positive values favour the first group named). n = 96 (Group A), 97 (Group B) and 99 (Group C).

Measure	Group A	Group B	Group C	A – B	A – C	C – B
Problem 1 (0–9)	6.09 (3.10)	2.95 (2.74)	4.89 (3.00)	1.07 [0.77, 1.37]	0.39 [0.11, 0.68]	0.67 [0.38, 0.96]
Problem 2a (0–7)	4.39 (2.95)	1.88 (2.80)	2.95 (3.15)	0.87 [0.57, 1.16]	0.47 [0.18, 0.75]	0.36 [0.08, 0.64]
Problem 2b: feasible improvements	1.27 (0.98)	0.09 (0.29)	1.07 (0.99)	1.63 [1.30, 1.95]	0.20 [–0.08, 0.48]	1.33 [1.02, 1.64]

Near transfer: Problem 1

On Problem 1, the difference between Groups A and B was large (g = 1.07), and Group C lay between them (Figure 5a). The curves of Groups A and C cross between 2 and 3 points: a higher proportion of Group C than of Group A reached the lowest thresholds (99% vs 94% scored at least 1 point, and 93% vs 85% at least 2 points), but fewer reached 3 points or more (63% vs 79%).

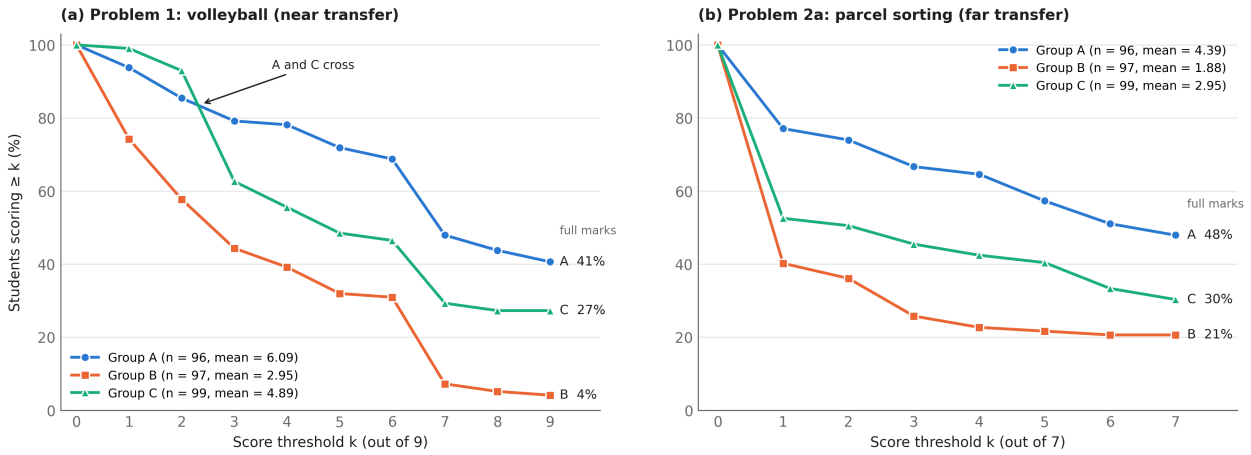


Figure 5. Percentage of students in each group scoring at least k points on (a) post-test Problem 1 (volleyball, near transfer; k = 0–9) and (b) Problem 2a (parcel sorting, far transfer; k = 0–7). The arrow in (a) marks where the curves of Groups A and C cross.

Far transfer: Problem 2

On Problem 2a, the gap between Groups A and C was wider than on Problem 1 (g = 0.47 vs 0.39; Figure 5b): Group C's mean score was 67% of Group A's, compared with 80% on Problem 1. Most of this difference came from how many students scored zero. About half of Group C scored zero on Problem 2a (47%, compared with 23% in Group A and 60% in Group B), whereas on Problem 1 almost every student in Group C earned at least one point.

On Problem 2b, Group C was close to Group A (Figure 6): 67% vs 80% proposed at least one feasible improvement, and 30% vs 33% proposed at least two. Among the students who scored zero on Problem 2a, 51% in Group C and 50% in Group A still proposed at least one feasible improvement. Group B proposed almost none: only 9% of its students proposed a feasible improvement, and 55% scored zero on both parts.

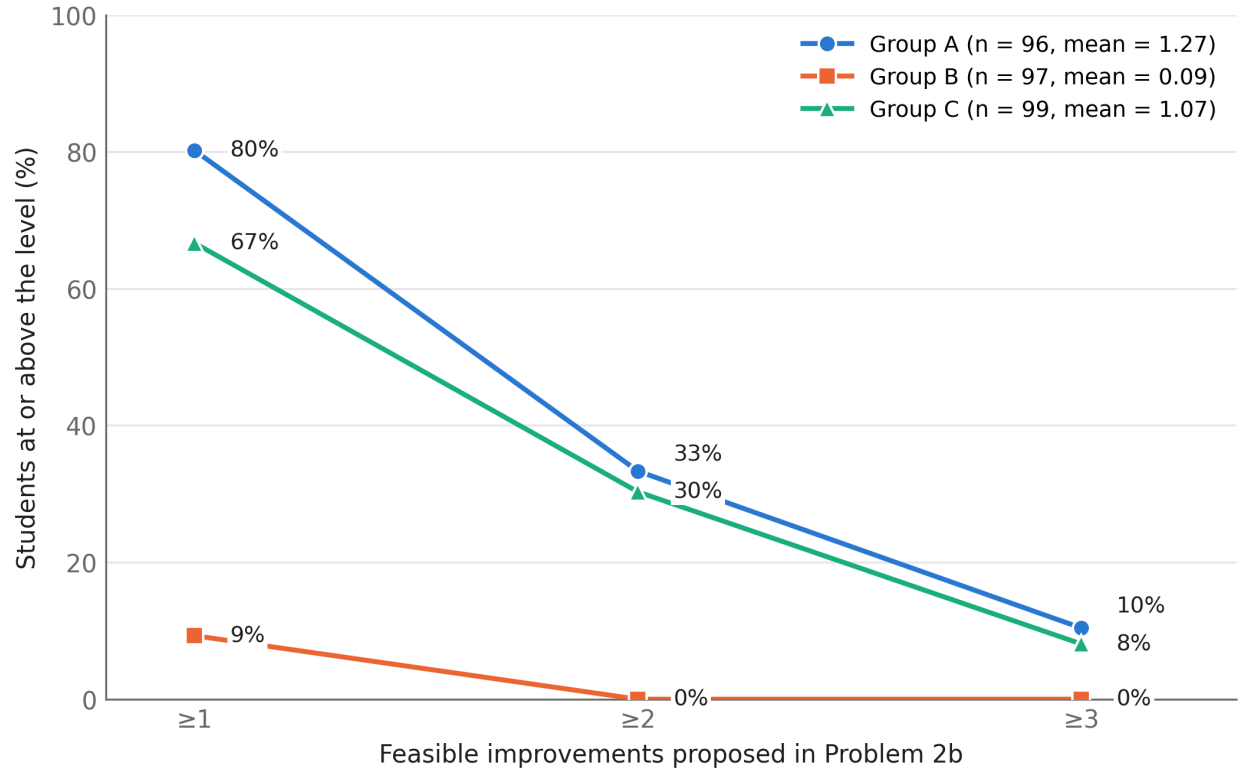


Figure 6. Post-test Problem 2b (parcel sorting): percentage of students in each group proposing at least one, two or three feasible improvements.

Written work during the homework

The written work collected from the homework differed sharply between the paper and the online groups. By the time they had checked their work against the answer, almost all students in Group A had written out a complete calculation on their homework sheets. The scratch paper of Groups B and C, by contrast, was mostly unused: 73% of the students in Group B and 61% in Group C handed in a blank sheet, and most of the others wrote only a few lines.

Trajectories drawn on the post-test

Figure 7 shows the trajectories that students drew on the answer sheets. Group C drew the limiting trajectories far more often than the other groups: 47% of Group C drew two trajectories on Problem 1 and 46% on Problem 2, compared with 3% and 10% in Group A and 11% on both problems in Group B. On Problem 1, 78% of Group C drew at least one trajectory, compared with 21% in Group A and 42% in Group B.

Drawing the limiting trajectories did not go together with higher scores. In Group C, students who drew two trajectories on Problem 1 scored lower on Problem 1 than those who drew none (mean 4.4 vs 6.7 out of 9, median 3 vs 6.5; g = −0.77, 95% CI [−1.29, −0.25]; p = 0.003): 55% of them scored 3 points or fewer, compared with 9% of those who drew none. On Problem 2a, students who drew two trajectories on Problem 2 scored about the same as those who drew none (mean 3.3 vs 2.9 out of 7; g = 0.14, 95% CI [−0.32, 0.61]; p = 0.65). In Groups A and B, too few students drew two trajectories for a meaningful comparison (between 3 and 11 students per problem).

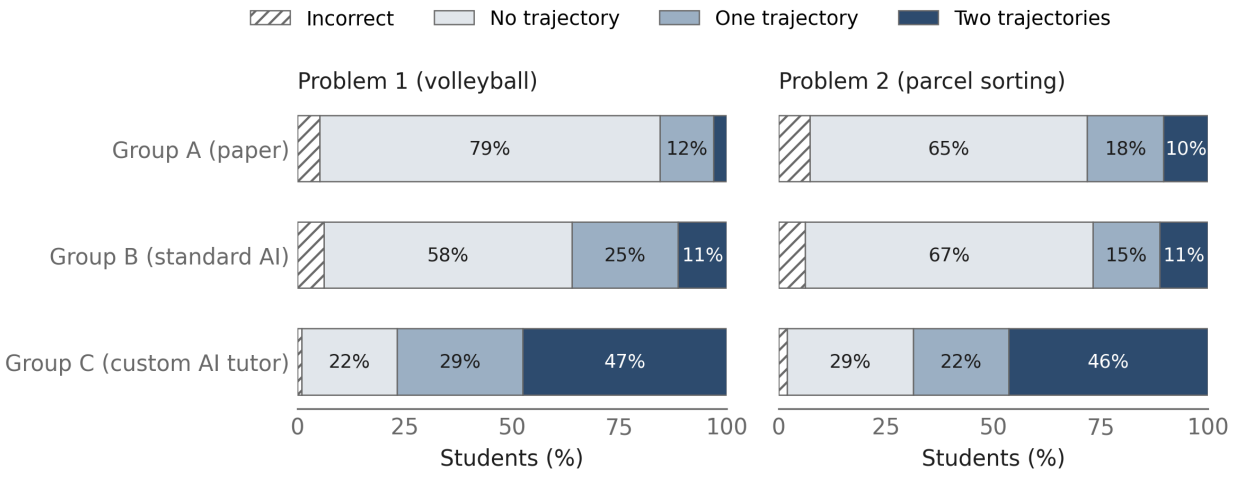


Figure 7. Trajectories drawn on the post-test answer sheets: percentage of students in each group who drew an incorrect trajectory, no trajectory, one trajectory or two trajectories (the two limiting trajectories), for Problem 1 (volleyball) and Problem 2 (parcel sorting).

Discussion

Overall pattern: paper > custom AI tutor > standard AI

The order of the groups follows how much of the work each homework format left to the students. Students in Group A had to think through and calculate every step themselves and then compare their work with a complete worked solution. This combination of practice and a worked solution is what research on worked examples recommends, particularly when learners explain the solution steps to themselves (Atkinson et al., 2000; Chi et al., 1989). Group C made decisions at a lower cost: most steps were choices among options, mistakes were corrected immediately, and students who were unsure could simply try an option, or ask the AI, instead of working the answer out carefully. Because mistakes were corrected at once, students had little reason to avoid them, and learners in interactive learning environments often do not use on-demand help effectively (Aleven et al., 2003). Students in Group B had the easiest path of all: they could ask the AI for help at any point. My review of the Group B chat records suggests that the AI readily gave students the help they asked for, often including the complete solution.

These results echo earlier findings. As in the study by Bastani et al. (2025), unrestricted access to a standard AI was followed by the weakest performance once the AI was taken away. In their study, the safeguards of a carefully prompted

tutor largely mitigated this harm; in this study, the custom AI tutor narrowed the gap, but Group C remained clearly behind Group A. The lower cost of each decision in Groups B and C also matches the finding of Stadler et al. (2024) that students using a large language model invested less mental effort and reasoned less deeply. Similarly, Fan et al. (2025) found that university students who revised essays with ChatGPT improved their essays the most but did not differ from other groups in knowledge gain or transfer. More generally, performance during practice is often an unreliable index of learning, and conditions that improve performance can fail to produce lasting learning (Soderstrom & Bjork, 2015).

Seeing the structure without calculating it

Group C's answer sheets suggest that many of these students knew what to do. Nearly half of them drew both limiting trajectories, far more than in the other groups, and they did so almost as often on Problem 2 as on Problem 1 (Figure 7). Students in Group C appear to have adopted the two limiting trajectories, which the scripted Coach and the AI Coach had pointed them to repeatedly, as their own way of representing the problem, and they used it in the new context as well. Representing a problem by its underlying structure rather than its surface features is characteristic of expertise (Chi et al., 1981). The drawings alone cannot show whether students understood these trajectories or simply reproduced what they had seen on the screen, but the results on Problem 2b, discussed below, point to at least some understanding. Group B worked with the same simulation but drew far less often, which suggests that this difference comes from the custom AI tutor rather than from the simulation itself. Identifying the limiting trajectories was not where they failed. This may also be why Group C did slightly better than Group A at the lowest score thresholds of Problem 1 (Figure 5a).

The drawings, however, did not turn into scores. Many students in Group C could not carry the solution through to the end. Those who drew two trajectories scored no better than those who drew none on Problem 2a, and lower on Problem 1, where nearly half of the Group C students who drew nothing earned full marks, compared with about a quarter of those who drew two trajectories (45% vs 23%). Drawing a representation does not by itself lead to a solution. Heckler (2010) found that introductory physics students who were prompted to draw a force diagram were less likely to solve the problem correctly than those who were not, whereas Rosengrant et al. (2009) found that students who drew diagrams correctly were more successful, and that high-achieving students used diagrams both to solve problems and to check their work.

One possible explanation is that being stuck led to the drawing, rather than the drawing leading to the answer: students who knew how to calculate may have gone straight to the solution, whereas students who did not know how to continue kept thinking about the problem and, in doing so, may have recalled the limiting trajectories that the Coach had shown them. During the homework, Group C typed only the whole-number quantities; other quantities, including the fall time and the boundary speed, were selected from options rather than calculated. Their scratch paper shows how little they wrote, and, in my observation, students in Groups B and C tended to watch what happened on the screen rather than work through the problem on paper. Most students in Group A, by contrast, wrote out the whole calculation on their homework sheets. Group C thus appears to have learned to see the structure of the problem but had little practice in calculating from it, and the gap between Groups A and C widened in the new context. This pattern is consistent with the assistance dilemma described by Koedinger and Aleven (2007): help that takes a step out of the learner's hands can make practice easier while reducing what the learner can later do alone.

Problem 2b, a qualitative question, gives a different picture. On this problem, Group C was close to Group A, and half of the Group C students who scored zero on Problem 2a still proposed a feasible improvement. These students had read the problem and understood qualitatively how a change in speed spreads the landing points, but they did not turn this understanding into a quantitative model. Group B differed from the other two groups: most of its students scored zero on both parts of Problem 2.

Implications

For the design of AI tutors, the results suggest that choosing from options should not replace calculation. A tutor could, for example, ask students to type in each calculated value, or to submit their written working, before it gives feedback. Kestin et al. (2025) showed that an AI tutor built on research-based teaching practices can outperform in-class active learning; the present results add that such a design should also leave the calculation to the student.

The results also raise a broader question. Students in Group A achieved the best scores, but in the age of AI, do students still need to be able to carry out the whole calculation themselves? An AI model has recently produced results on several hundred open problems in mathematics (OpenAI, 2026), which makes it more pressing to ask what students most need to learn. As Figure 7 shows, Group C had already formed an overall picture of the problem and of the trajectories involved. A scoring system that rewards complete calculation may not capture everything that students learn from AI-supported homework.

In informal conversations after the homework, students in Group C said that they felt confident they had learned the topic and that they enjoyed this way of studying, and some students in Groups B and C asked for the link so that they could use the page again at home. Such impressions should be interpreted with caution, because people often mistake their performance during practice for learning (Soderstrom & Bjork, 2015). Whether this interest turns into a lasting motivation to learn remains to be studied.

Limitations

This was a quasi-experiment: each condition was given to two whole classes rather than to randomly assigned students, and group equivalence was established on general physics examinations rather than on a pretest of projectile motion.

Group A also differed from Groups B and C in ways other than the presence of AI. Group A completed the homework on paper during an evening self-study session, whereas Groups B and C completed it on a computer within a fixed 20-minute slot in the computer room; time on task was not measured. Group A had a complete written solution to study after attempting the problem, whereas Group B had no solution and Group C encountered the solution only step by step through the tutor. The post-test required students to write out complete calculations on paper, which matched the practice of Group A but not that of Groups B and C, who rarely wrote during the homework. Groups B and C were also using the online page for the first time, so part of their homework time may have been spent learning the interface. In addition, the standard AI may sometimes have given incorrect solutions; this reflects how students use AI on their own, but it may have contributed to Group B's low scores.

Finally, the study covered a single topic, a single homework task and one post-test.

Conclusion

This study compared three homework formats for the same horizontal projectile motion problem. Students who did traditional paper homework performed best on both a near-transfer and a far-transfer post-test problem, students who used a standard AI performed worst, and students who used a custom AI tutor were in between. The custom AI tutor appears to have taught students to see the structure of the problem: they drew the two limiting trajectories far more often than the other groups, including in a new context, and they reasoned about the new context qualitatively almost as well as the paper group. However, it did not prepare them to carry out the calculation on their own. As long as physics teaching and assessment expect students to carry out complete calculations on their own, AI tutors for physics homework should therefore be designed so that students still do the calculations themselves. Whether students still need this ability in the age of AI remains an open question.

Declarations

Ethics statement

The homework and the post-test were part of regular teaching and did not count towards students' grades. Students were informed that anonymised answer sheets might be published, and each student could choose not to have their answer sheet published. The answer sheets of the 29 students who chose not to (Group A 13, Group B 12, Group C 4) are not included in the supplementary material. All students are identified only by anonymous IDs, assigned in random order within each group.

Competing interests

The author developed the online homework page and the custom AI tutor used in this study, and declares no financial competing interests.

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Data availability

The anonymised post-test scores and trajectory codes of all 292 students, and the blurred answer sheets of the 263 students who consented to publication, are available at https://phylab.uk/research/Click_or_Calculate/. The prompts used for the standard AI and the custom AI tutor are given in Appendices A and B, which are provided as a separate file at the same address, and an English demonstration version of the online homework page is available at https://phylab.uk/labs/volleyball-serve/.

Use of AI in preparing this manuscript

During the preparation of this manuscript, the author used Claude (Anthropic) to improve the English language, to help draft and restructure parts of the text, to help identify and check references, and to write the scripts used for the data analysis and figures. The author reviewed and edited all content and takes full responsibility for the manuscript.

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